Software Requirements Analysis Document

Project: 42_DigiConnect — Unified Citizen Service Delivery Platform (UCSDP)

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Compliant with Scott W. Ambler's "The Elements of UML 2.0 Style" and Roger S. Pressman's Software Engineering (7th Ed.)

1. Executive Summary & System Overview

DigiConnect is an enterprise-grade digital governance platform designed to eliminate bureaucratic bottlenecks, physical paperwork, and opaque delays in citizen service delivery. Inspired by national e-governance platforms (*Digital India*, *MeeSeva*, *Seva Sindhu*, *e-District*), DigiConnect introduces an automated end-to-end framework featuring:

- **Decentralized Bottom-Up Territorial Routing:** Automatically directs citizen applications to localized Village/Ward officers based on self-referential jurisdiction trees.
- **Dynamic Multi-Step Workflow Engine:** Empowers Department Heads to design custom form fields, mandatory proofs, and sequential officer designations rather than hardcoding static administrative titles.
- **Algorithmic SLA Enforcement:** Monitors statutory review countdowns, freezes timers during citizen clarification periods, and automatically reassigns overdue files up the jurisdiction tree.
- **Closed-Loop Grievance Redressal:** An independent dispute mechanism with legal authority to uphold, direct re-verification, or overrule rejections and issue digital certificates directly.
- **Multi-Tier Financial Segregation:** Automatically segregates transaction revenue into statutory Department Service Fees and Central Platform Maintenance Fees.

2. Administrative Domain & 6-Tier Hierarchy

Public administration in DigiConnect is structured on an adjacency list hierarchical tree reflecting constitutional territorial boundaries:

```
Tier 1: Central Government (National Tenant Oversight)
   └── Tier 2: State Government (e.g., Andhra Pradesh, Karnataka)
         └── Tier 3: Revenue Division / Zone
               └── Tier 4: District (e.g., Tirupati, Chittoor)
                     ├── Tier 5 (Rural): Mandal (e.g., Chandragiri)
                     │     └── Tier 6 (Rural): Village / Gram Panchayat (e.g., Chandragiri Village)
                     └── Tier 5 (Urban): Municipality / Municipal Corporation (e.g., Tirupati MC)
                           └── Tier 6 (Urban): Ward (e.g., Ward 14)
```

- **Bottom-Up Routing Principle:** Applications submitted by citizens selecting Tier 6 nodes land directly in the queue of the L1 officer stationed at that specific node (e.g., Village Revenue Officer or Ward Clerk).
- **Hierarchical Oversight:** Higher-tier officials have scoped read and supervisory reassignment authority across all child nodes under their jurisdiction branch.

3. User Roles & Stakeholder Specification

The system supports 6 primary roles:

Role ID	Role Name	Scope & Authority	Primary Responsibilities
`CITIZEN`	Registered Citizen	Citizen Portal	Discover services, submit dynamic applications, upload proofs, pay fees, track live SLAs, download digital certificates, file grievances.
`OFFICER`	Field / Review Officer	Local Jurisdiction Node	Review assigned applications, inspect uploaded proofs, perform 3 actions: Approve & Advance, Reject with grounds, or Raise Query.
`DEPARTMENT_HEAD`	Department Administrator	Department-wide	Create custom role designations, configure dynamic services (form fields, proofs, workflow steps), monitor department revenue.

Role ID	Role Name	Scope & Authority	Primary Responsibilities
GRIEVANCE_OFFICER	Grievance Authority	Grievance Cell / District	Investigate appeals, review rejection histories, execute one of three legally binding resolutions: Uphold, Direct Re-verification, or Overrule.
STATE_ADMIN	State Executive	State-wide	Manage state jurisdiction tree (add/edit districts, mandals, wards), onboard departments and heads, track state SLA metrics, suspend officers/departments.
CENTRAL_ADMIN	Platform Super-User	National	Onboard states (1:1 mapping with State Admins), inspect national analytics, audit global system ledger, configure platform parameters, deactivate state tenants.

4. Functional Requirements (FRs)

4.1 Module 1: Citizen Services Portal

- **Authentication & Profile:** The system shall allow citizens to authenticate using Email, Phone, or Aadhaar number, and securely maintain their residential jurisdiction profile.
- **Service Discovery:** The system shall present a searchable catalog of public services categorized by Department (Revenue, Municipal, Social Welfare, Civil Supplies, Civil Registration).
- **Dynamic Form Rendering:** When a citizen selects a service, the system shall dynamically generate form fields, validations, and document upload requirements defined by the Department Head.
- **Jurisdiction Selection:** The system shall require the citizen to select their local jurisdiction via a cascading tree dropdown (State $\rightarrow$ District $\rightarrow$ Mandal/Municipality $\rightarrow$ Village/Ward).
- **Fee Payment & Receipting:** The system shall integrate with an External Payment Gateway boundary to collect the combined Service Fee and Platform Fee, generating a cryptographically verifiable payment receipt.
- **Application Tracking & Timeline:** The system shall provide real-time tracking for each application (`APP-YYYY-XXXX`), displaying the active step, assigned officer designation, remaining SLA hours, and query alerts.
- **Digital Certificate Download:** Upon final sanction, the system shall provide instant download of the PDF digital certificate (`CERT-YYYY-XXXX`) complete with digital signature metadata and tamper-evident QR verification code.

4.2 Module 2: Field Officer Verification & Dynamic Multi-Step Workflow

- **Jurisdiction Queue Retrieval:** The system shall display to the logged-in officer only applications whose selected jurisdiction falls within the officer's administrative boundary.
- **Split-Screen Document Verification:** The review interface shall present citizen form details alongside uploaded document attachments with zoom, rotate, and document validation toggles.
- **Action 1 — Approve & Advance:**
  - If the current step is an intermediate step, the system shall advance the application to the next step's designated officer.
  - If the current step is the designated final approval step, the system shall mark the application `COMPLETED`, generate the digital certificate, and dispatch issuance notifications.
- **Action 2 — Reject:**
  - The officer must provide mandatory statutory rejection grounds.

  - The system shall lock the application status to `REJECTED`, record the entry in the immutable timeline, and send rejection grounds to the citizen with appeal instructions.
- **Action 3 — Raise Query:**
  - The officer may request document re-upload or clarification.
  - The system shall transition the application status to `QUERY_RAISED` and notify the citizen.
  - **The SLA countdown clock shall immediately pause**, preventing penalty accrual against the officer until the citizen responds.

4.3 Module 3: Dynamic SLA & Algorithmic Auto-Reassignment

- **Statutory Timer Initiation:** Upon confirmed payment and submission, the system shall initialize an SLA due date based on the service's configured statutory SLA days.
- **Query Pause/Resume Cycle:** When an officer raises a query, the SLA timer must freeze. When the citizen re-submits requested documents, the SLA timer resumes with the remaining hours.
- **Algorithmic Auto-Reassignment upon Breach:**
  - When the SLA countdown reaches zero without officer action, the system shall mark the state as `ESCALATED`.
  - The system shall automatically traverse the jurisdiction tree upward to identify the immediate parent supervisor (e.g., Mandal Revenue Officer / Tahsildar).
  - The application shall be automatically reassigned to the parent supervisor's queue with priority status `CRITICAL`.
  - An automated escalation audit record and email/SMS alert shall be generated.

4.4 Module 4: Department Head Dynamic Service & Workflow Builder

- **Role Designation Management:** The Department Head shall create custom role designations (e.g., `VRO`, `Revenue Inspector`, `Tahsildar`, `Town Planning Officer`) dynamically per department.
- **Dynamic Service Builder:** The Department Head shall configure Service Name, Service Code, Department Category, Statutory Service Fee, Central Platform Fee, and Statutory SLA threshold.
- **Dynamic Form & Proof Customizer:** The Department Head shall configure custom fields (Text, Number, Date, Dropdown, Constraints) and specify mandatory document attachments.
- **Multi-Step Workflow Modeler:** The Department Head shall specify sequential workflow steps:
  - Step Number, Step Name.
  - Required Officer Designation for that step.
  - Permitted Actions: `canApprove`, `canReject`, `canRaiseQuery`.
  - `isFinalApprovalStep` toggle (designating the final sanctioning officer who triggers certificate generation).

4.5 Module 5: Closed-Loop Grievance Redressal

- **Grievance Filing:** A citizen can file a grievance against any rejected application or prolonged SLA delay, providing grounds and supplementary proof.
- **Quasi-Judicial Action A — Uphold Rejection:** Grievance Officer confirms that the field rejection was statutory; marks grievance `RESOLVED`, leaving the application `REJECTED`.
- **Quasi-Judicial Action B — Direct Re-Verification:**
  - Grievance Officer determines that the applicant has submitted valid new evidence.
  - The system reopens the original application to `PENDING_REVERIFICATION` and reassigns it to the designated supervisor.
  - The historical audit trail retains both the original rejection and grievance reopening.
- **Quasi-Judicial Action C — Overrule & Issue Certificate:**
  - Grievance Officer determines that field rejection was arbitrary or unlawful.

  - The system overrules the field rejection, sets the application status to `COMPLETED`, generates the digital certificate immediately, and stamps the Grievance Officer as the issuing authority.

4.6 Module 6: State Administration, Jurisdiction Tree & Suspension Governance

- **Jurisdiction Tree Builder:** State Admin shall create, edit, or reorganize territorial nodes across Rural (District $\rightarrow$ Mandal $\rightarrow$ Village) and Urban (District $\rightarrow$ Municipality $\rightarrow$ Ward) divisions.
- **Department & Office Onboarding:** State Admin onboards statutory state departments and assigns Department Heads.
- **Suspension Governance:** State Admin shall have statutory authority to suspend non-compliant Department Officers or freeze Department operational queues pending departmental inquiry.
- **State-Wide Governance Dashboard:** Real-time dashboard showing SLA breach rates, average clearance hours, and department-wise grievance resolution compliance.

4.7 Module 7: Central Government, Tenant Lifecycle & Deactivation Control

- **State Tenant Lifecycle:** Central Admin creates and manages State entities, enforcing 1:1 mapping with verified State Admins.
- **Tenant Deactivation Governance:** Central Admin shall possess the executive capability to deactivate non-compliant state tenants or revoke state administrative authorizations.
- **National Application Analytics:** Centralized analytics aggregating total applications, clearance ratios, and SLA compliance nationwide.
- **National Revenue Oversight:** Aggregates national platform fee collections and department service disbursements.

4.8 Module 8: Financial Management & Multi-Tier Fee Segregation

- **Fee Segregation Model:** For every service payment:

  $$\text{Total Fee} = \text{Service Fee (Payable to Department)} + \text{Platform Fee (Payable to Central Infrastructure)}$$

- **Department Revenue Reconciliation:** Department Heads can view itemized revenue reports detailing service fee earnings per service and transaction.
- **State & Central Financial Rollup:**
  - State Admin views department breakdown and accumulated state platform fee share.
  - Central Admin views national totals with automated reconciliation ledgers.

4.9 Module 9: Tamper-Evident Audit Ledger

- **Immutable Action Logging:** Every state change, query raise, citizen response, approval, rejection, and reassignment shall append an immutable record to the application's timeline containing:
  - Timestamp (ISO 8601 UTC)
  - Actor ID, Actor Name, Actor Designation
  - Action Category & Formal Remarks
- **System Traceability:** Audit records cannot be modified or deleted by any user, including Central Administrators.

5. Non-Functional Requirements (NFRs)

5.1 Performance & Throughput

- **Response Time:** API responses for dashboard metrics, queue queries, and state transitions shall execute within ≤ 500 ms under normal load.

- **Concurrency:** The system architecture shall support at least 1,000 concurrent citizen submissions and 200 active officer reviews without degradation.
- **Certificate Generation Time:** Digital certificate generation and PDF cryptographic stamping shall take ≤ 2 seconds from final approval trigger.

5.2 Security & Role-Based Access Control

- **Role-Based Access Control - RBAC:** Strict verification of user authorization tokens and normalized roles (`CITIZEN`, `OFFICER`, `DEPARTMENT_HEAD`, `GRIEVANCE_OFFICER`, `STATE_ADMIN`, `CENTRAL_ADMIN`) on all API endpoints.
- **Data Sanitization & Injection Prevention:** Strict validation of incoming payloads using NestJS `ValidationPipe`, `class-validator`, and sanitized parameterized data models.
- **Document Privacy:** Citizen proof documents shall be stored in access-restricted object directories accessible only to authorized reviewing officers within the jurisdiction chain.

5.3 Reliability, Availability & Fault Tolerance

- **Availability:** The platform shall maintain 99.9% uptime during operational service delivery hours.
- **Stateless API Design:** The backend REST services shall remain stateless, propagating identity headers to allow seamless horizontal load balancing.
- **Graceful API Fallbacks:** Frontend components shall implement resilient fallback states (e.g., `Promise.all` with localized `.catch()` fallbacks) so an error in secondary widgets does not crash primary workflows.

5.4 Usability & Accessibility

- **Responsive Design:** The application interface shall be fully responsive across desktop (1920×1080), tablet (768×1024), and mobile (375×667) form factors.
- **Keyboard & Screen-Reader Accessibility:** All critical dashboard interactive items and queue cards shall feature appropriate ARIA roles, tabindex attributes, and keyboard navigation support (`Enter` / `Space`).

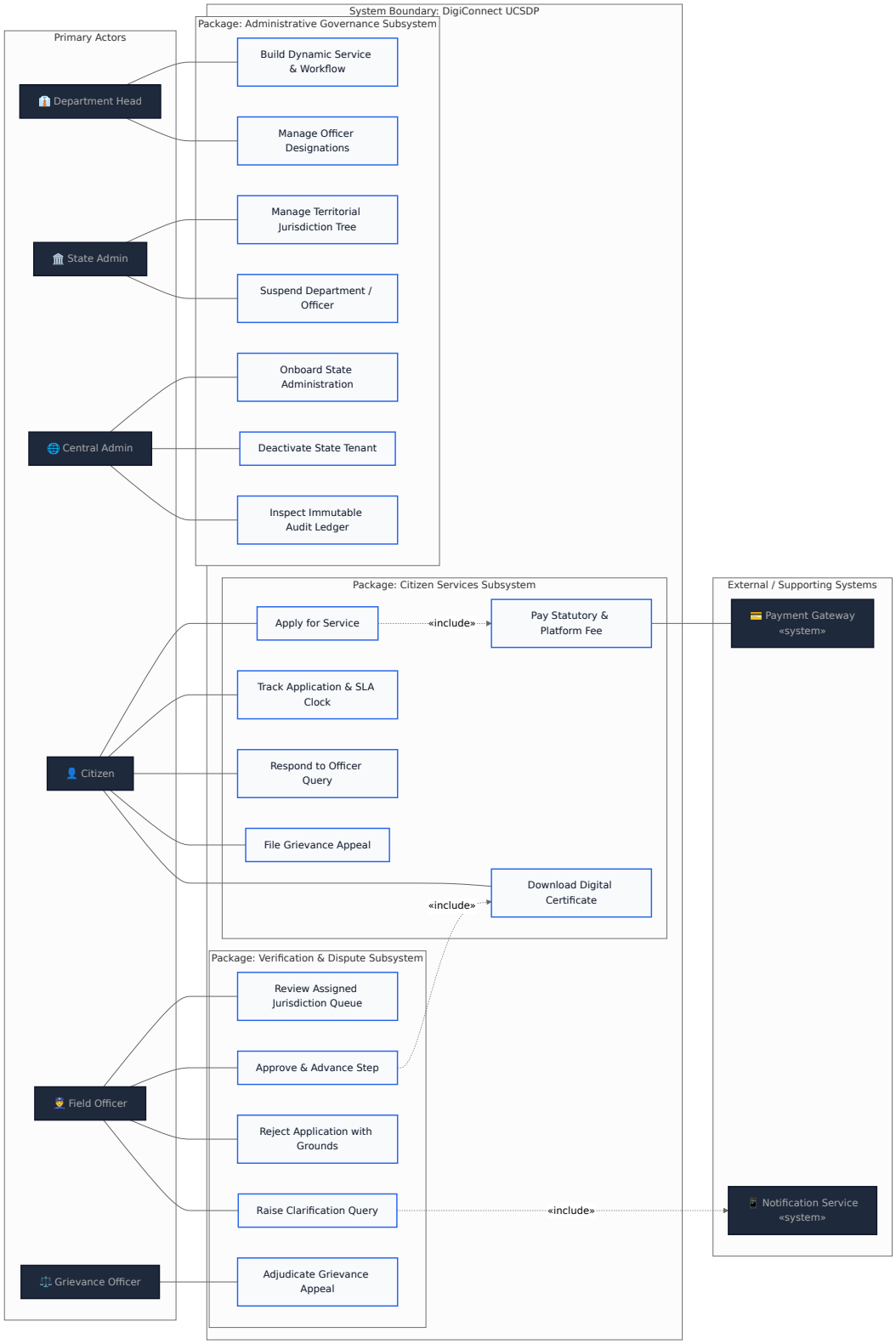
5.5 Maintainability, Portability & Relational Schema Readiness

- **Relational Migration Readiness:** The domain data models (`User`, `Service`, `Application`, `JurisdictionNode`, `Grievance`, `Certificate`) shall adhere to 3NF relational normalization with explicit foreign keys.
- **In-Memory Prototype Compatibility:** For testing and educational demonstration, the system executes on a deterministic in-memory store (`store.ts` & `bottom-up-seeder.ts`) with zero external database dependencies.

6. UML Diagrams

6.1 Unified System Use-Case Diagram (with Subsystem Packages)

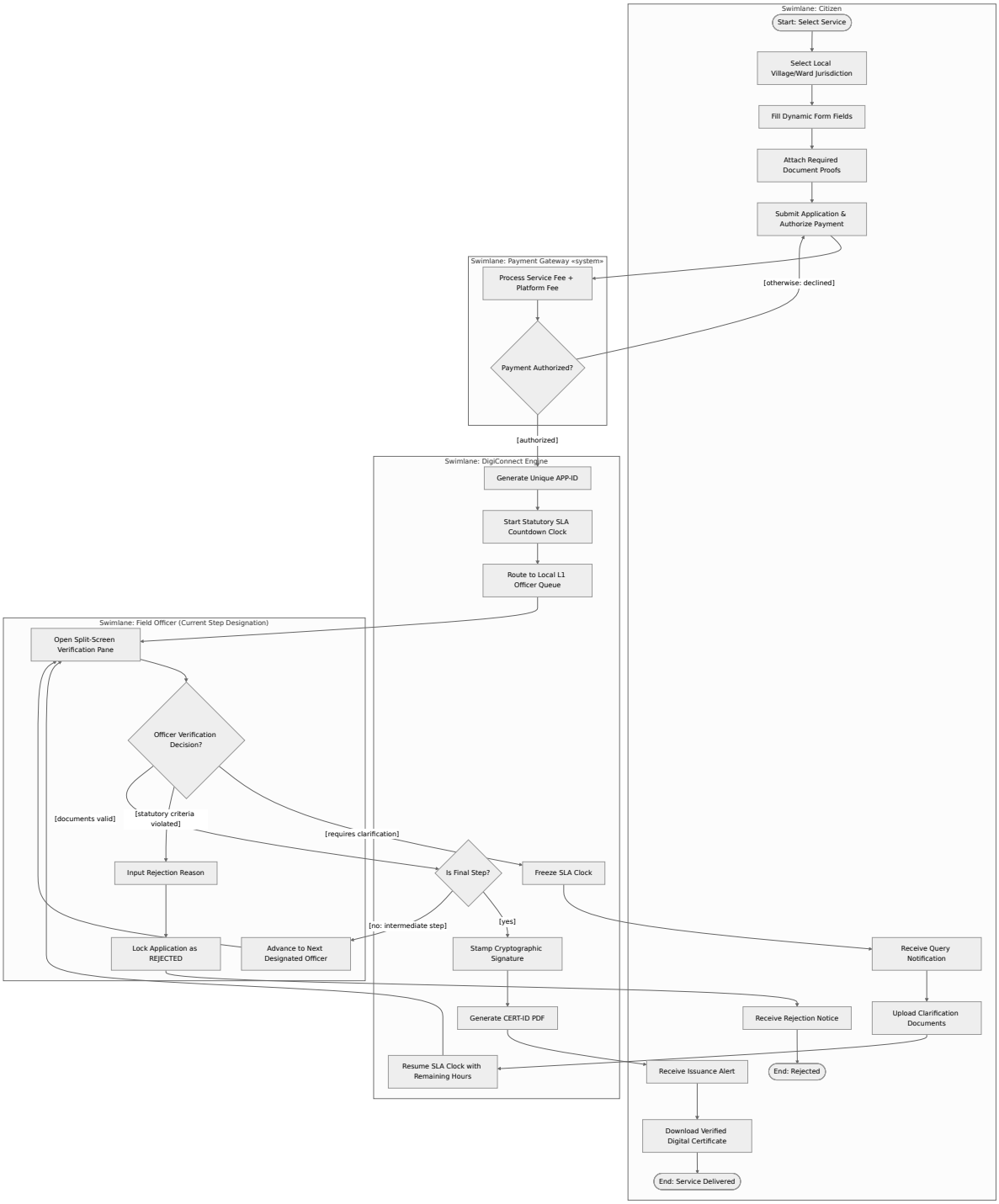
Compliant with Ambler Rule 16, 61, 62, 66, 70, 71, 72, 85, & 150. Proactive actors are on the left; reactive system actors on the right; solid lines for associations without arrows; structured into 3 clear Subsystem Packages.



6.2 Process-Oriented Activity Diagrams (with Swimlanes)

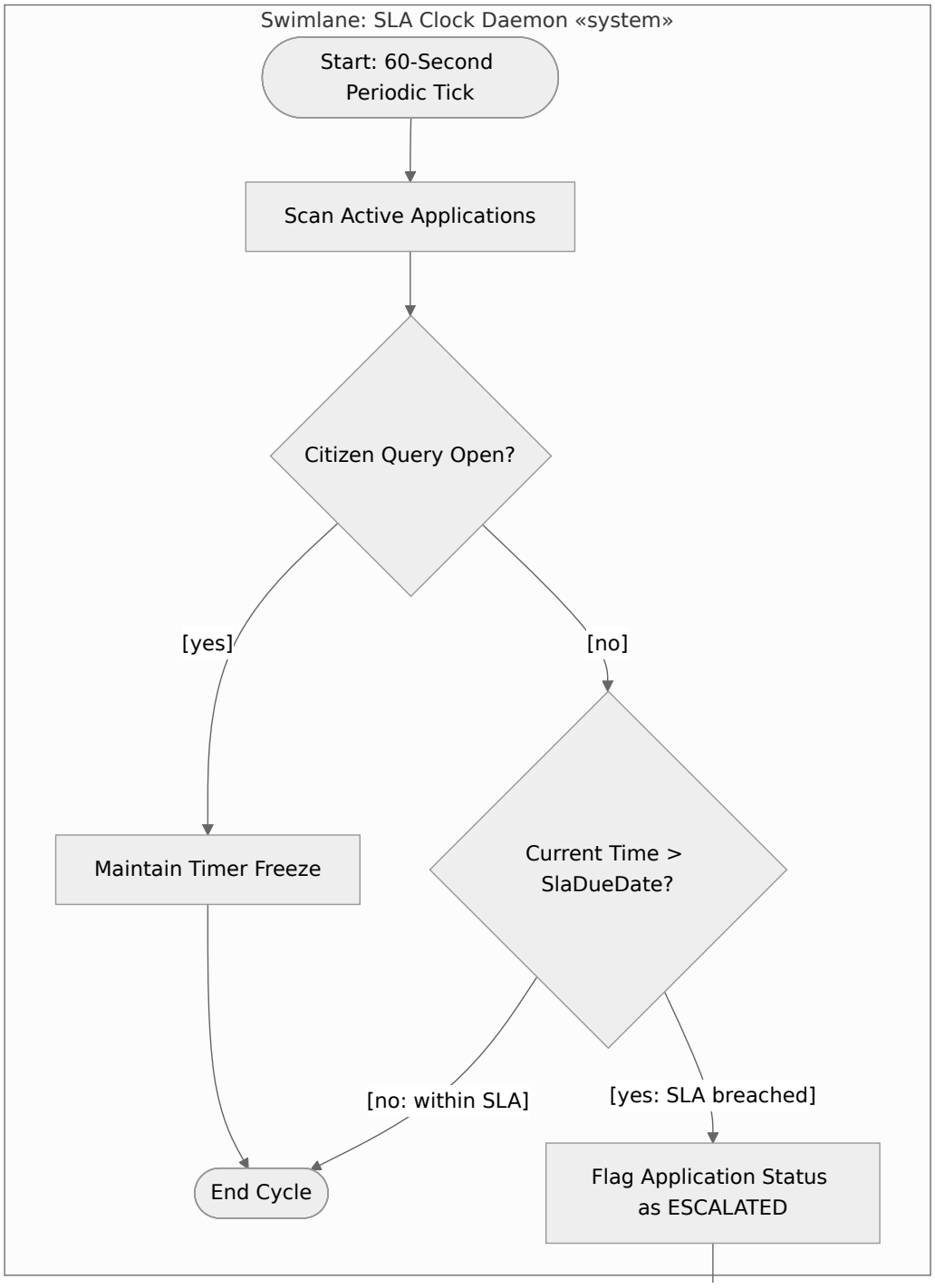
Activity Diagram 1: Citizen Application, Payment & Multi-Step Verification

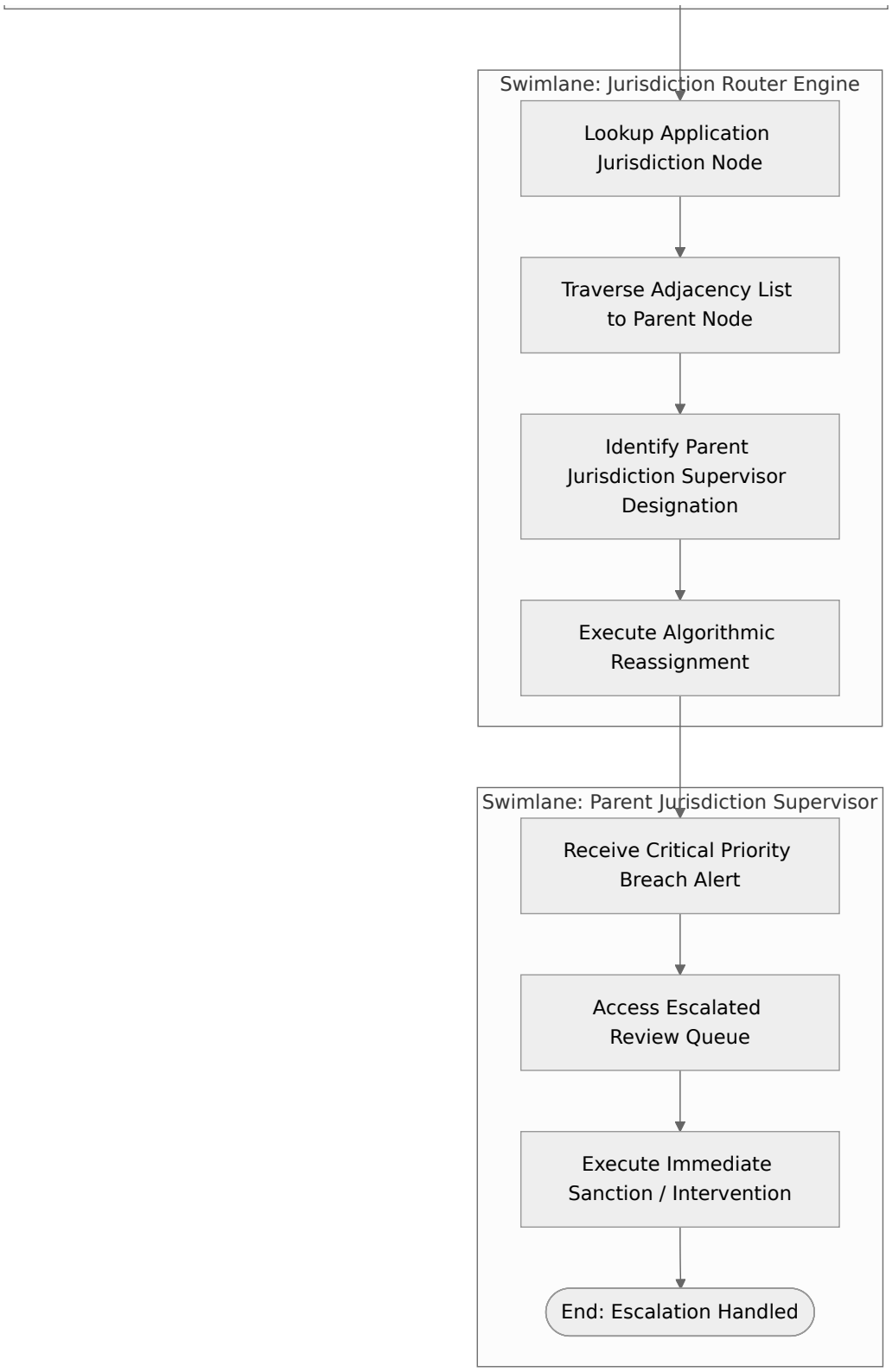
Compliant with Ambler Chapter 10: Horizontal Swimlanes, less than 5 partitions (Rule 248), explicit decision points with mutually exclusive guards (Rule 231, 232).



Activity Diagram 2: SLA Clock Daemon & Algorithmic Upward Reassignment

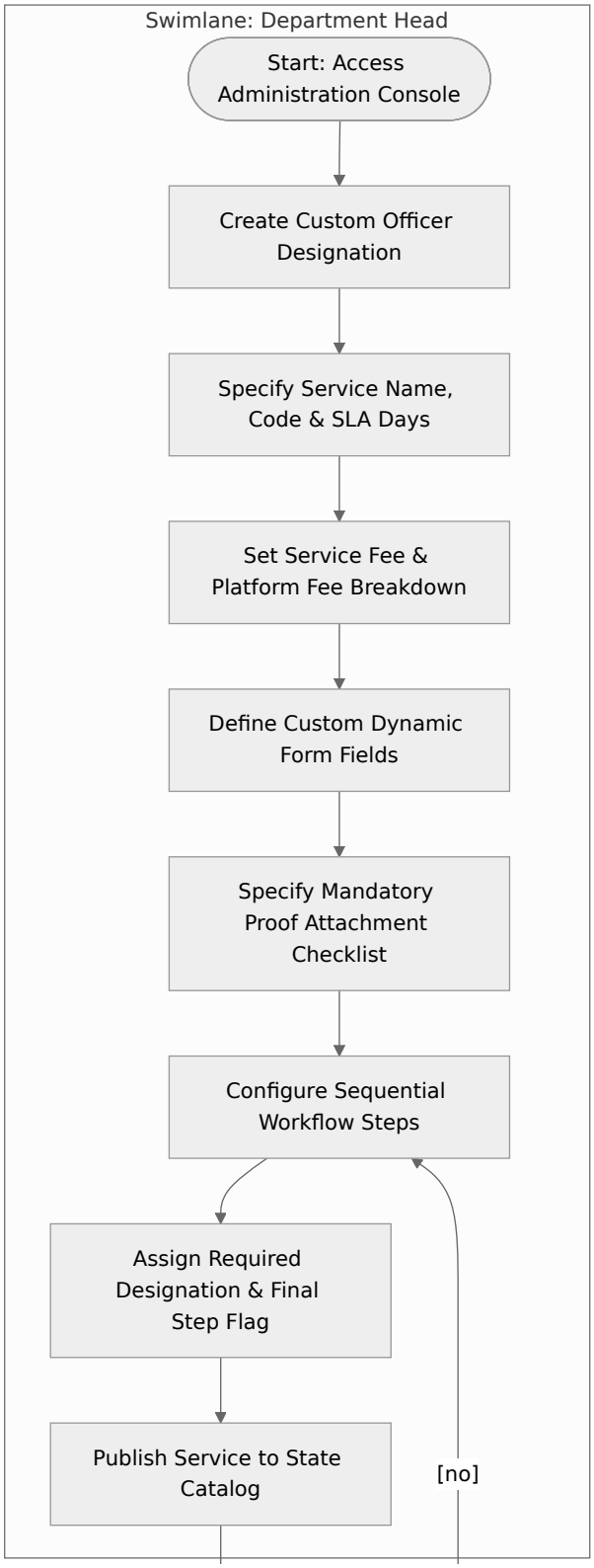
Compliant with Ambler Rules 221, 222, 231, 233, and 247.

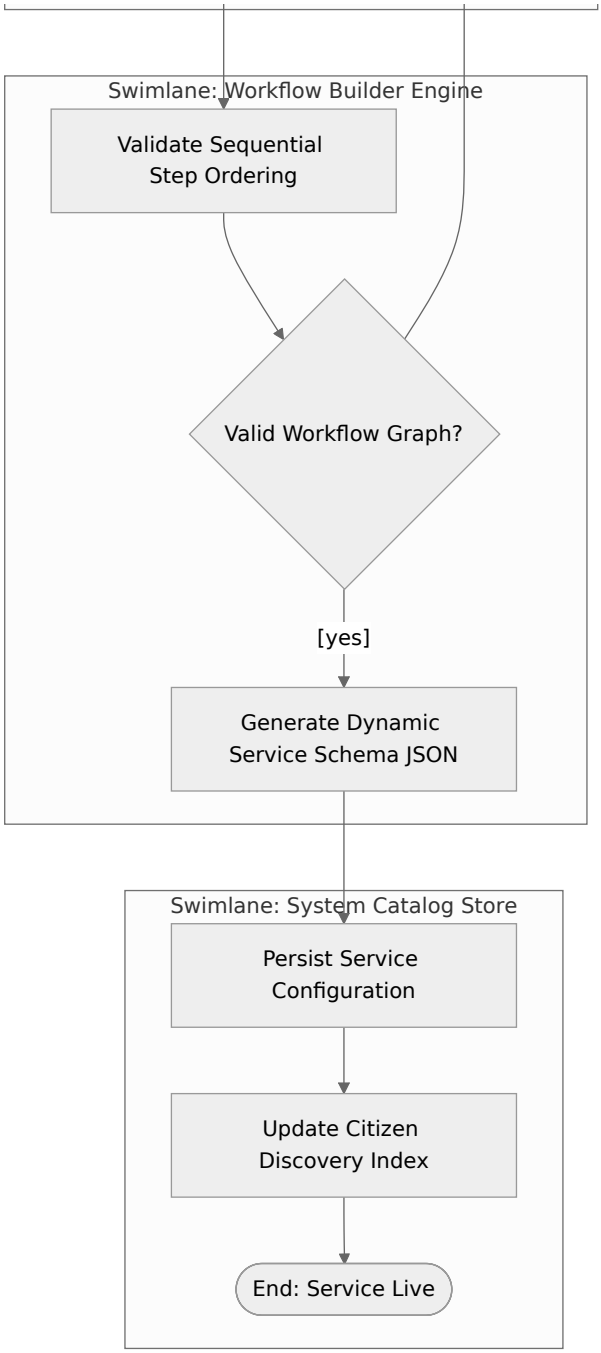



Swimlane: Jurisdiction Router Engine
Lookup Application Jurisdiction Node
Traverse Adjacency List to Parent Node
Identify Parent Jurisdiction Supervisor Designation
Execute Algorithmic Reassignment
Swimlane: Parent Jurisdiction Supervisor
Receive Critical Priority Breach Alert
Access Escalated Review Queue
Execute Immediate Sanction / Intervention
End: Escalation Handled

Activity Diagram 3: Department Head Dynamic Service & Workflow Builder

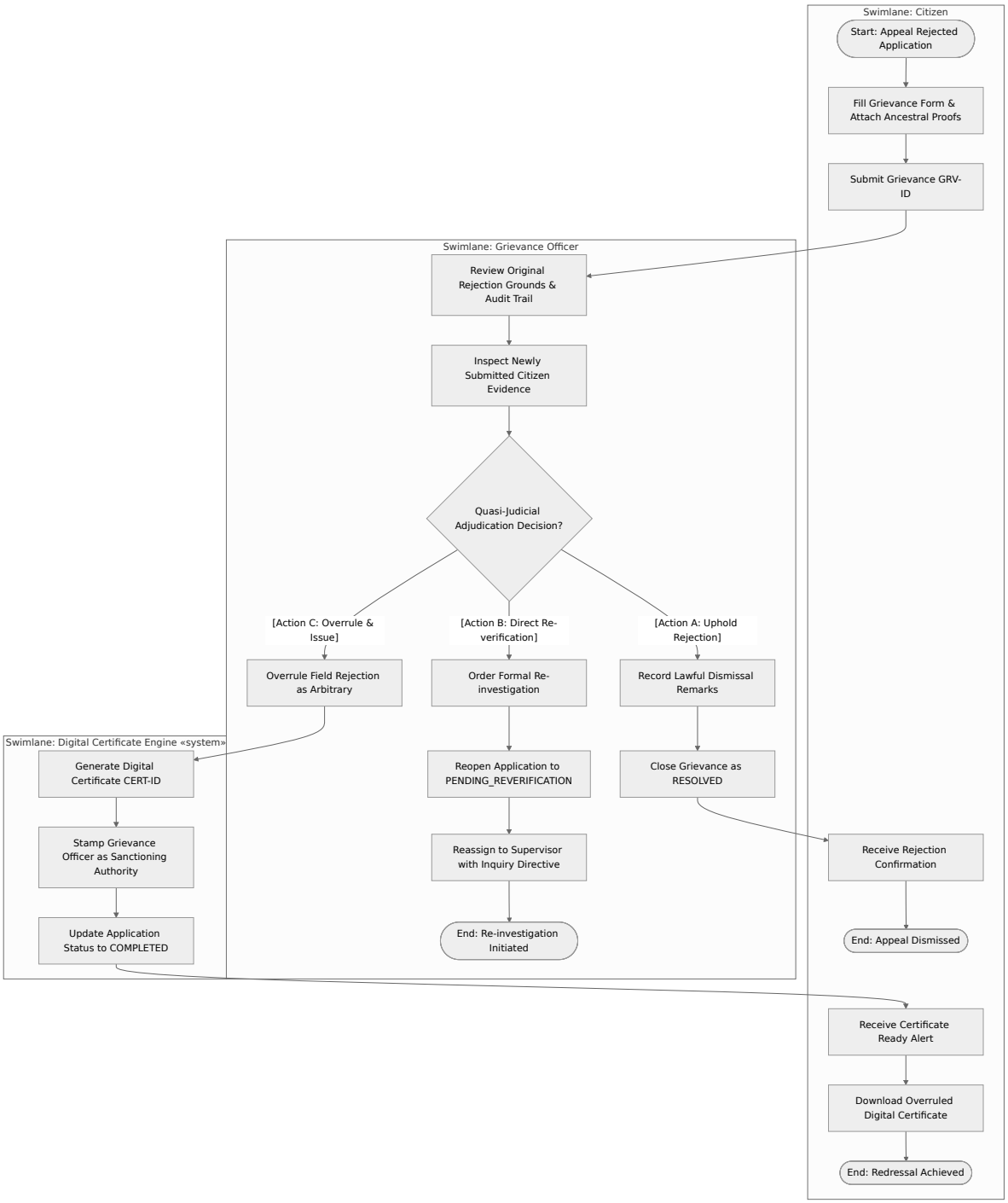
Models dynamic creation of officer designations and multi-stage schemas.




Swimlane: Workflow Builder Engine
Validate Sequential Step Ordering
Valid Workflow Graph?
[yes]
Generate Dynamic Service Schema JSON
Swimlane: System Catalog Store
Persist Service Configuration
Update Citizen Discovery Index
End: Service Live

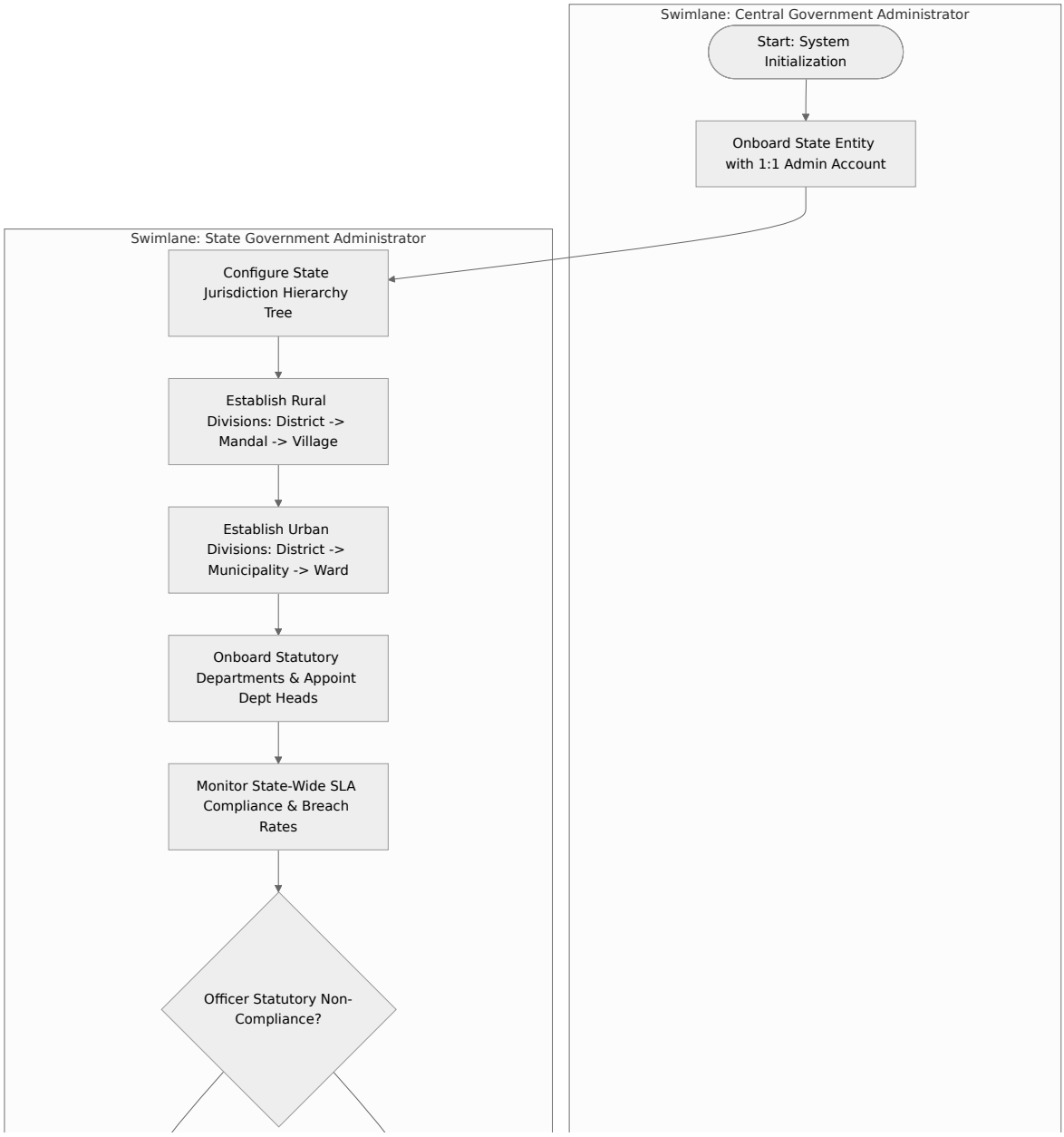
Activity Diagram 4: Closed-Loop Grievance Investigation & Adjudication

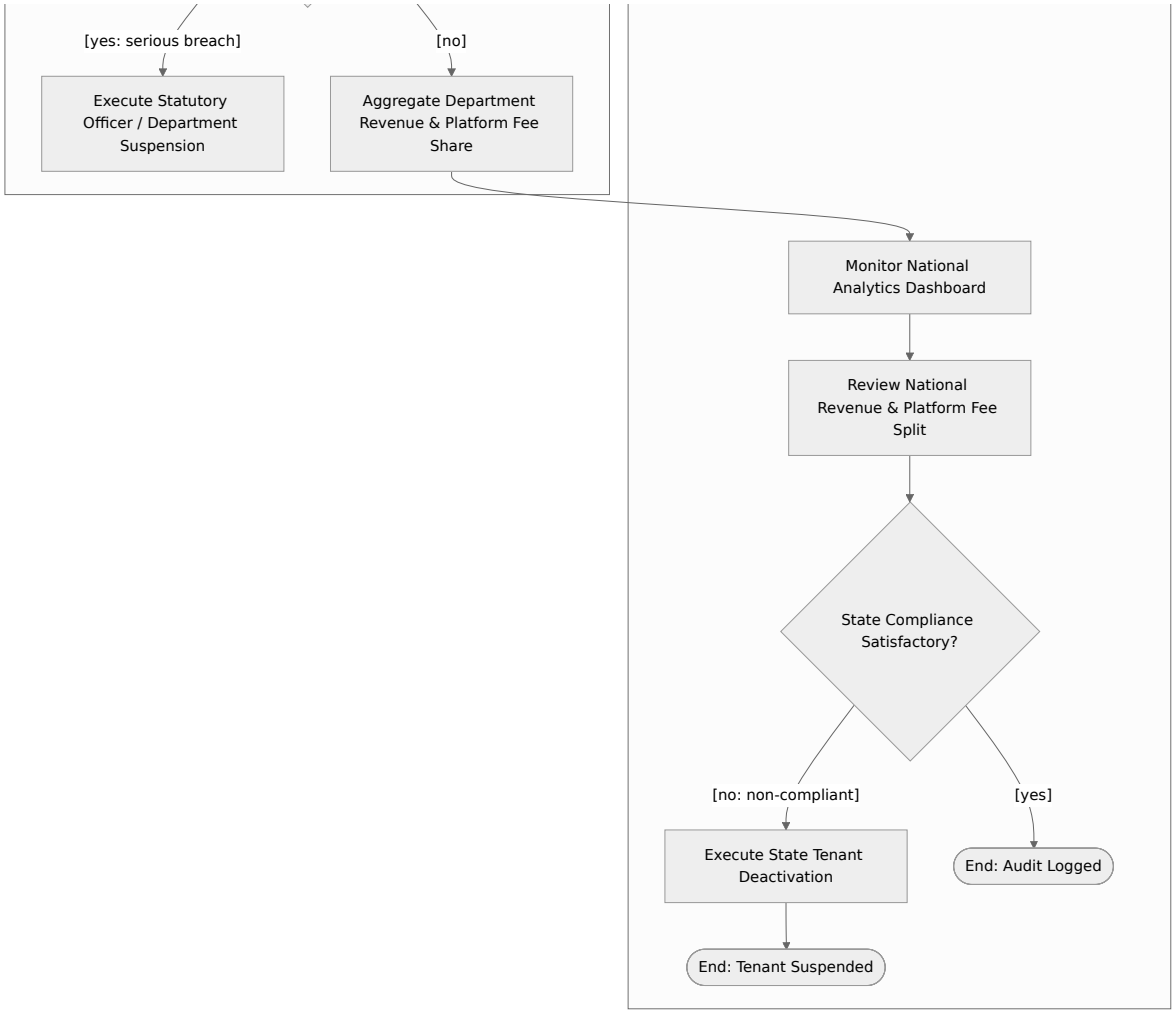
Captures the 3 legally binding quasi-judicial resolutions.



Activity Diagram 5: State & Central Multi-Tier Administrative Governance

Encompasses tenant onboarding, territorial tree setup, suspension, and financial reconciliation.



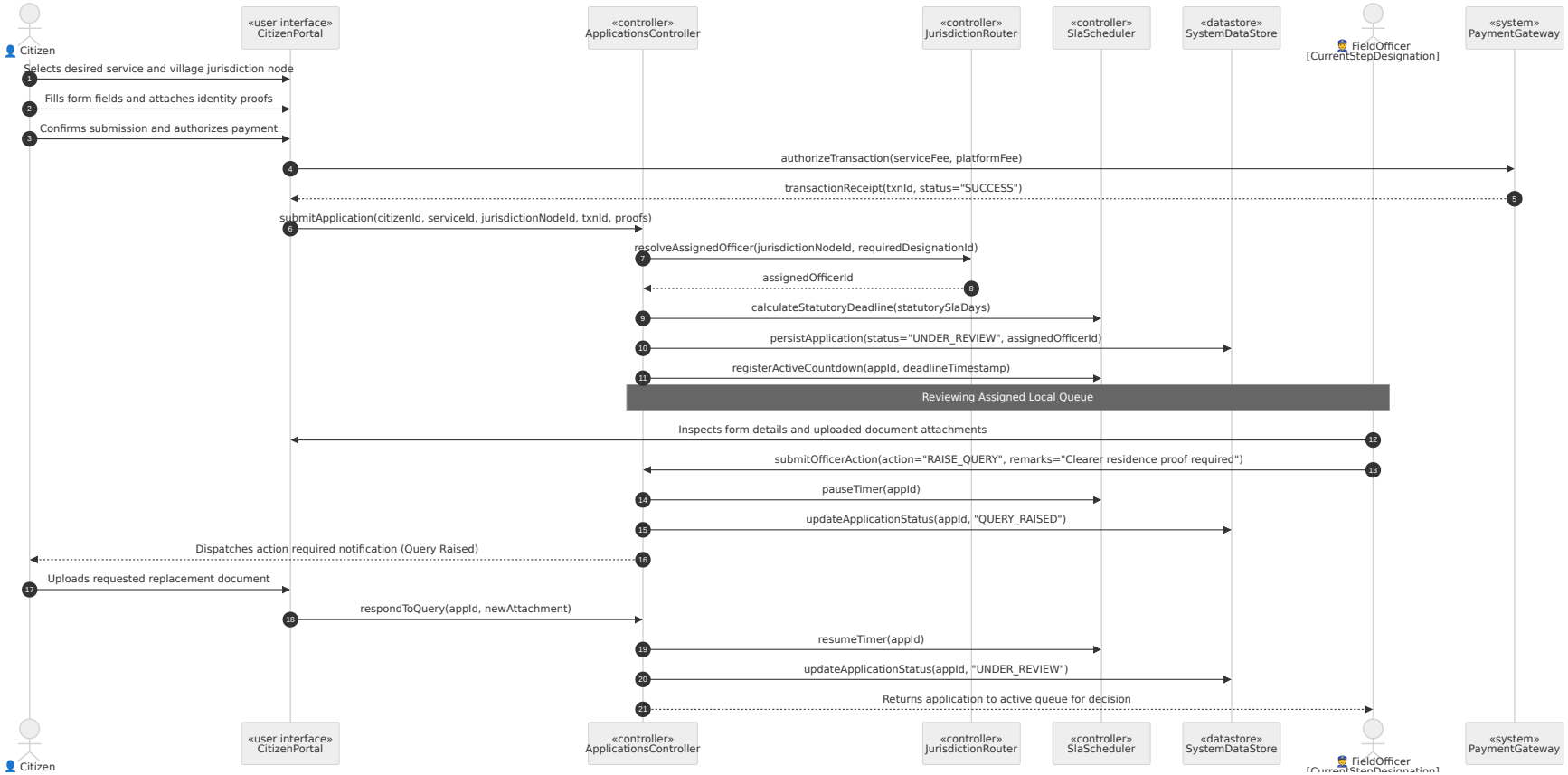


6.3 Layered Sequence Diagrams (with Dynamic Lifelines)

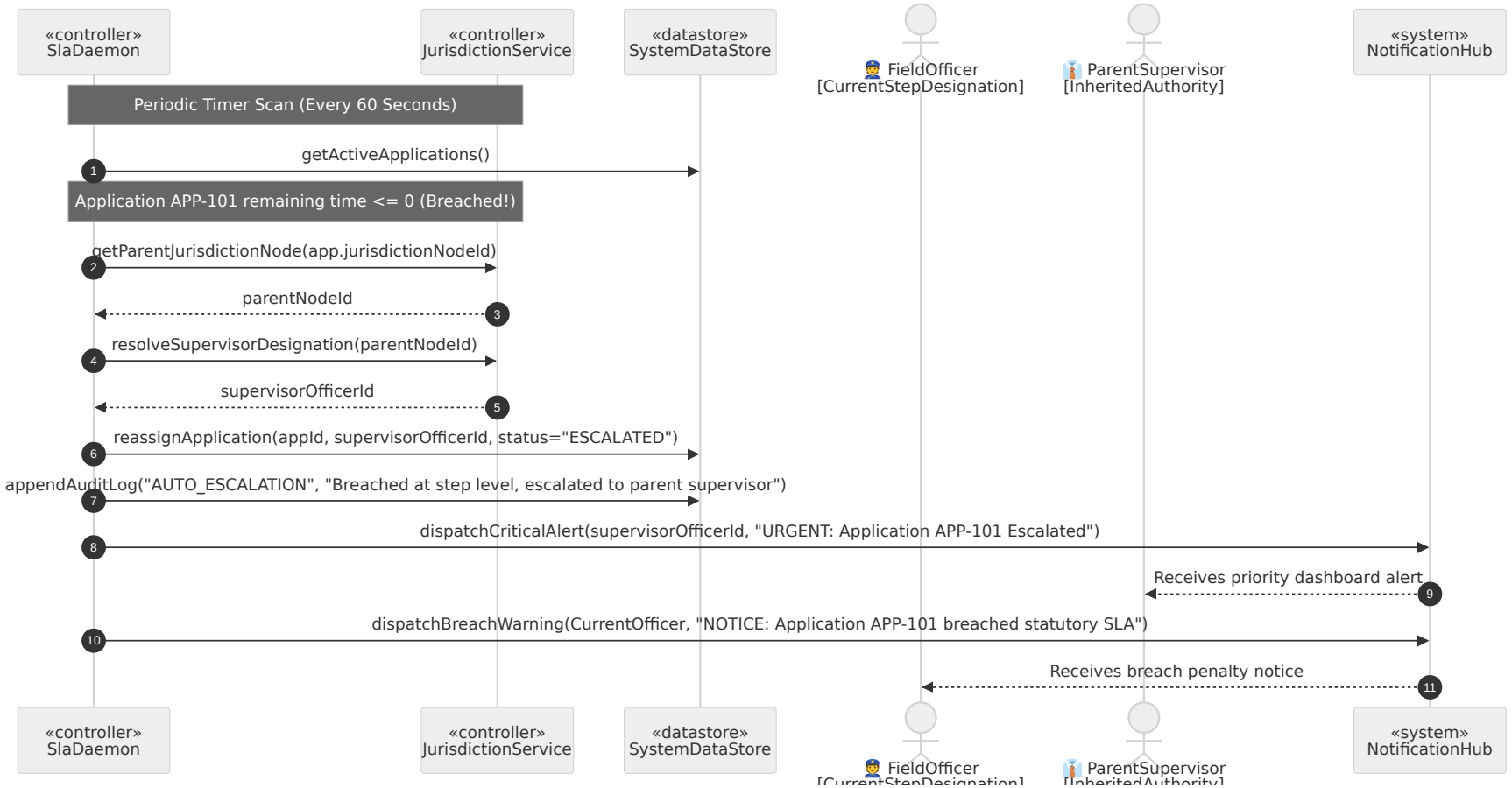
Compliant with Ambler Chapter 7:

- **Rule 159 & 163 (Layering):** Actor (leftmost) → User Interface → Controller → DataStore → External System (rightmost).
- **Rule 173 (Actor Messages):** Prose for human interactions; operation signatures for software messages (Rule 172).
- **Rule 168 (Stereotypes):** Textual stereotypes applied consistently.
- **Rule 177:** Obvious return arrows omitted.
- **Rule 65:** Lifelines model dynamic roles (`CurrentStepOfficer`, `ParentSupervisor`), **not hard-coded job titles like VRO or Tahsildar**.

Sequence Diagram 1: Dynamic Application Submission, Verification & SLA Query Pause



Sequence Diagram 2: Dynamic Algorithmic SLA Breach & Upward Reassignment



Sequence Diagram 3: Grievance Overrule & Direct Certificate Issuance

